الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني
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الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2011 الاولى

الاختصاص والرمز : الالكترونيك (26204)
المادة : الكترونيك رقمي
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المعدل: (10)
المدة: ساعتان

Treat only four questions: (5 pts/question)

I) **a)** Quote the addressing modes of the following instructions of a ***68HC11***:

a) LSLA
b) BPL DELAY
c) LDY #$00FA
d) STX $04, X
e) LDD #$0039
f) STAA $0001

b) Explain in details the following ***expressions***:

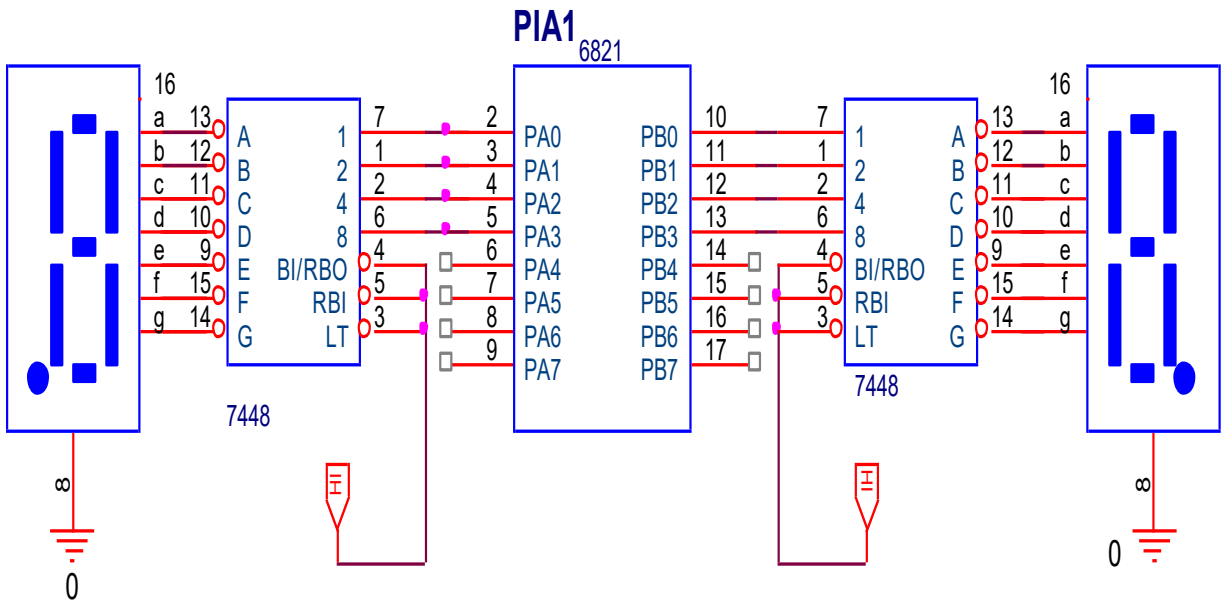
1. Interrupt Vector
2. IRQ
3. Index Register X, Y
4. CCR
5. Stack Pointer
6. SCI

c) Write the program that initializes the A/D converter of the ***68HC11*** in ***signal mode*** at PE3, at continuous conversion and displays the binary value that results from the conversion to ***PORTC***.

II) **a)** Explain in PIC ***16F84A*** the following instructions:

a. BTFSC STATUS, C
b. GOTO DELAY
c. DECFSZ NUM, W
d. SUBWF COUNT, F
e. ROLF NUMBER, W
f. MOVWF PORTB

b) Using the ***6809 MPU***, two BCD decoders connected to ports A and B of a PIA and two 7 segment displays; write a program to build an odd counter from 01 to 89.



PIA Registers:

ORA et DDRA = $8000	CRA = $8001
ORB et DDRB = $8002	CRB = $8003

III) A ***68HC11*** *MCU* is connected to :

- 8 switches (S_0, S_1, S_2, S_3, S_4, S_5, S_6 and S_7)
- 8 Leds (L_0, L_1, L_2, L_3, L_4, L_5, L_6 and L_7)
- a push button K.

The LEDs are connected to ***PORT B***, the switches are connected to ***PORT E*** and the push button is connected to ***IRQ*** interrupt pin.

a- Draw the simple connection circuit that shows the ***MCU***, the Leds (cathode to GND) and the switches (Switch ON=Switch connected to Ground) and the push button.

b- Write the necessary program that reads the value at ***PORT E***, when an interrupt occurs at ***IRQ*** and displays the result at ***PORT B***.

* Interrupt vector of ***IRQ*** is $FFF2-$FFF3.

IV) **a)** Explain the role of the registers ***SCCR1***, ***SCCR2*** and ***SCSR*** for the initialization of the asynchronous serial interface ***SCI***:

SCCR1 — SCI Control Register 1 **$102C**

Bit 7	6	5	4	3	2	1	Bit 0
R8	T8	—	M	WAKE	—	—	—

SCCR2 — SCI Control Register 2 **$102D**

Bit 7	6	5	4	3	2	1	Bit 0
TIE	TCIE	RIE	ILIE	TE	RE	RWU	SBK

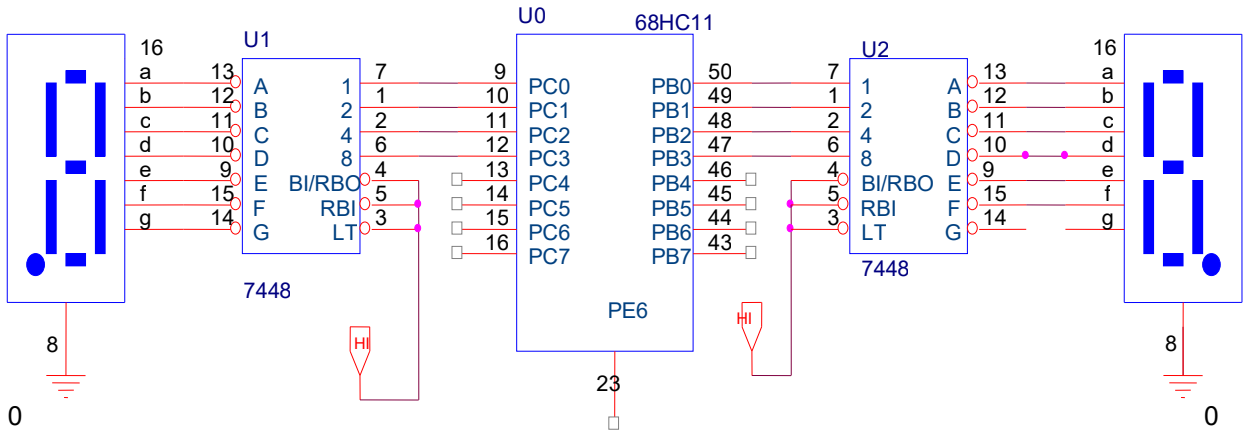
SCSR — SCI Status Register **$102E**

Bit 7	6	5	4	3	2	1	Bit 0
TDRE	TC	RDRF	IDLE	OR	NF	FE	—

b) Write in ***68HC11*** a program that allows to sequentially output to ***PORTC***, the 8 outputs of the table below:

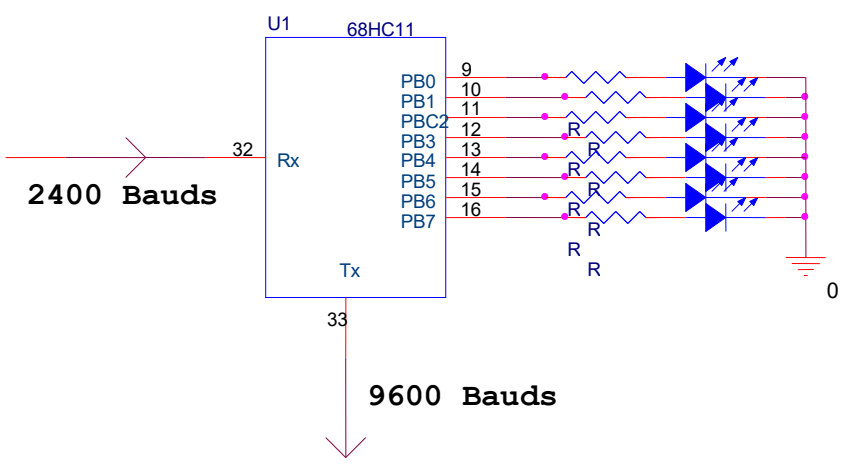
PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
0	0	1	1	1	1	1	1
1	0	0	1	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	0	0	1	1	1
1	1	1	1	0	0	1	1
1	1	1	1	1	0	0	1
1	1	1	1	1	1	0	0
1	1	1	1	1	1	0	0
1	1	1	1	1	0	0	1
1	1	1	1	0	0	1	1
1	1	1	0	0	1	1	1
1	1	0	0	1	1	1	1
1	0	0	1	1	1	1	1
0	0	1	1	1	1	1	1

V) **a)** Write the program that achieves according to the below principle, an up counter at ***PORTC*** and a down counter at ***PORTB***, every time that ***PE6*** goes to ***"0"***. Each going to ***"0"*** achieves a complete counting.



b) Explain the operation and the registers of the ***Timer*** of the ***68HC11*** in the ***Input Capture*** mode.

VI) Consider the following circuit:



A ***68HC11*** receives a series of 74 values of 8 bits ***by interruption*** under ***2400*** bauds and stores them in a table that starts by address $***00A0***. Once the table filled, the microcontroller changes the baud rate to ***9600*** and retransmits them one after another, through the ***SCI without interruption*** and by examining the bit ***TDRE*** of the ***SCCR2*** register. At the same time that a value is transmitted, it should be displayed at ***PORT B***. The quartz of the ***68HC11*** is of ***8 MHZ***.
Write the program that achieves what precede.

HC11 INSTRUCTION SET:

ABA	ABX	ABY	ADCA	ADCB	ADDA	ADDB	ADDD	ANDA
ANDB	ASL	ASLA	ASLD	ASR	ASRA	ASRB	BCC	BCLR
BCS	BEQ	BGE	BGT	BHI	BHS	BITA	BITB	BLE
BLS	BLT	BMI	BNE	BPL	BRA	BRCLR	BRN	BRSET
BSR	BVC	BVS	CBA	CLC	CLI	CLR	CLRSA	CLRB
CLV	CMPA	CMPB	CPD	CPX	CPY	COM	COMA	COMB
DAA	DEC	DECA	DECB	DES	DEX	DEY	EORA	EORB
FDIV	IDIV	INC	INCA	INCB	INS	INX	INY	JMP
JSR	LDAA	LDAB	LDD	LDS	LDX	LDY	LSL	LSLA
LSLB	LSLD	LSLX	LSR	LSRA	LSRB	LSRD	MUL	NEG
NEGA	NEGB	NOP	ORAA	ORAB	PSHA	PSHB	PULA	PULB
PULB	PSHX	PSHY	PULX	PULY	ROL	ROLA	ROLB	ROR
RORA	RORB	RTI	RTS	SBA	SBCA	SBCB	SEC	SEI
SEV	STAA	STAB	STD	STOP	STS	STX	STY	SUBA
SUBB	SUBD	SWI	TAB	TAP	TBA	TPA	TST	TSTA
TSTB	TSX	TSY	TXS	TYS	WAI	XGDX	XGDY	

INTERRUPT	INTERRUPT VECTOR
SCI	$FFD6 ; $FFD7
IRQ	$FFF2 ; $FFF3

$1000	PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0	PORTA
$1001	DDA7	DDA6	DDA5	DDA4	DDA3	DDA2	DDA1	DDA0	DDRA
$1002	PG7	PG6	PG5	PG4	PG3	PG2	PG1	PG0	PORTG
$1003	DDG7	DDG6	DDG5	DDG4	DDG3	DDG2	DDG1	DDG0	DDRG
$1004	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0	PORTB
$1005	PF7	PF6	PF5	PF4	PF3	PF2	PF1	PF0	PORTF
$1006	PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0	PORTC
$1007	DDC7	DDC6	DDC5	DDC4	DDC3	DDC2	DDC1	DDC0	DDRC
$1008	0	0	PD5	PD4	PD3	PD2	PD1	PD0	PORTD
$1009	0	0	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
$100A	PE7	PE6	PE5	PE4	PE3	PE2	PE1	PE0	PORTE
$100B	FOC1	FOC2	FOC3	FOC4	FOC5	0	0	0	CFORC
$100C	OC1M7	OC1M6	OC1M5	OC1M4	OC1M3	0	0	0	OC1M
$100D	OC1D7	OC1D6	OC1D5	OC1D4	OC1D3	0	0	0	OC1D
$100E	Bit 15	14	13	12	11	10	9	Bit 8	TCNT (High)
$100F	Bit 7	6	5	4	3	2	1	Bit 0	TCNT (Low)
$1010	Bit 15	14	13	12	11	10	9	Bit 8	TIC1 (High)
$1011	Bit 7	6	5	4	3	2	1	Bit 0	TIC1 (Low)
$1012	Bit 15	14	13	12	11	10	9	Bit 8	TIC2 (High)
$1013	Bit 7	6	5	4	3	2	1	Bit 0	TIC2 (Low)
$1014	Bit 15	14	13	12	11	10	9	Bit 8	TIC3 (High)
$1015	Bit 7	6	5	4	3	2	1	Bit 0	TIC3 (Low)
$1016	Bit 15	14	13	12	11	10	9	Bit 8	TOC1 (High)
$1017	Bit 7	6	5	4	3	2	1	Bit 0	TOC1 (Low)
$1018	Bit 15	14	13	12	11	10	9	Bit 8	TOC2 (High)
$1019	Bit 7	6	5	4	3	2	1	Bit 0	TOC2 (Low)
$101A	Bit 15	14	13	12	11	10	9	Bit 8	TOC3 (High)
$101B	Bit 7	6	5	4	3	2	1	Bit 0	TOC3 (Low)
$101C	Bit 15	14	13	12	11	10	9	Bit 8	TOC4 (High)
$101D	Bit 7	6	5	4	3	2	1	Bit 0	TOC4 (Low)
$101E	Bit 15	14	13	12	11	10	9	Bit 8	TI4/O5 (High)
$101F	Bit 7	6	5	4	3	2	1	Bit 0	TI4/O5 (Low)
$1020	OM2	OL2	OM3	OL3	OM4	OL4	OM5	OL5	TCTL1
$1021	EDG4B	EDG4A	EDG1B	EDG1A	EDG2B	EDG2A	EDG3B	EDG3A	TCTL2
$1022	OC1I	OC2I	OC3I	OC4I	I4/O5I	IC1I	IC2I	IC3I	TMSK1
$1023	OC1F	OC2F	OC3F	OC4F	I4/O5F	IC1F	IC2F	IC3F	TFLG1
$1024	TOI	RTII	PAOVI	PAII	0	0	PR1	PR0	TMSK2
$1025	TOF	RTIF	PAOVF	PAIF	0	0	0	0	TFLG2
$1026	0	PAEN	PAMOD	PEDGE	0	I4/O5	RTR1	RTR0	PACTL
$1027	Bit 7	6	5	4	3	2	1	Bit 0	PACNT

$1028	SPIE	SPE	DWOM	MSTR	CPOL	CPHA	SPR1	SPR0	SPCR
$1029	SPIF	WCOL	0	MODF	0	0	0	Bit 0	SPSR
$102A	Bit 7	6	5	4	3	2	1	Bit 0	SPDR
$102B	TCLR	0	SCP1	SCP0	RCKB	SCR2	SCR1	SCR0	BAUD
$102C	R8	T8	0	M	WAKE	0	0	0	SCCR1
$102D	TIE	TCIE	RIE	ILIE	TE	RE	RWU	SBK	SCCR2
$102E	TDRE	TC	RDRF	IDLE	OR	NF	FE	0	SCSR
$102F	Bit 7	6	5	4	3	2	1	Bit 0	SCDR
$1030	CCF	0	SCAN	MULT	CD	CC	CB	CA	ADCTL
$1031	Bit 7	6	5	4	3	2	1	Bit 0	ADR1
$1032	Bit 7	6	5	4	3	2	1	Bit 0	ADR2
$1033	Bit 7	6	5	4	3	2	1	Bit 0	ADR3
$1034	Bit 7	6	5	4	3	2	1	Bit 0	ADR4
$1035	0	0	0	PTCON	BPRT3	BPRT2	BPRT1	BPRT0	BPROT
$1036									Reserved
$1037									Reserved
$1038	GWOM	CWOM	CLK4X	0	0	0	0	0	OPT2
$1039	ADPU	CSEL	IRQE	DLY	CME	FCME	CR1	CR0	OPTION
$103A	Bit 7	6	5	4	3	2	1	Bit 0	COPRST
$103B	ODD	EVEN	0	BYTE	ROW	ERASE	EELAT	EEPGM	PPROG
$103C	RBOOT	SMOD	MDA	IRV	PSEL3	PSEL2	PSEL1	PSEL0	HPRIO
$103D	RAM3	RAM2	RAM1	RAM0	REG3	REG2	REG1	REG0	INIT
$103E	TILOP	0	OCCR	CBYP	DISR	FCM	FCOP	0	TEST1
$103F	EE3`	EE2	EE1	EE0	1	NOCOP	1	EEON	CONFIG

SCP1	SCP0	Division Factor	Crystal Frequency				
			2^{23} Hz	8 MHz	4.9152 MHz	4 MHz	3.6864 MH
			Highest Baud Rate				
0	0	1	131.072K Baud	125.000K Baud	76.80K Baud	62.50K Baud	57.60K Baud
0	1	3	43.691K Baud	41.667K Baud	25.60K Baud	20.833K Baud	**19.20K Baud**
1	0	4	32.768K Baud	31.250K Baud	19.20K Baud	15.625K Baud	14.40K Baud
1	1	13	10.082K Baud	9600 Baud	5.908K Baud	4800 Baud	4431K Baud
			2.1 MHz	2 MHz	1.2288 MHz	1 MHz	921.6 kHz
			Bus Frequency (E clock)				

SCR 2	SCR 1	SCR 0	Divi-sion Factor	Highest Baud Rate (from Table 9-1)				
				131.072K Baud	32.768K Baud	76.80K Baud	19.20K Baud	9600 Baud
				SCI Baud Rate				
0	0	0	1	131.072K Baud	32.768K Baud	76.80K Baud	19.20K Baud	9600 Baud
0	0	1	2	65.536K Baud	16.384K Baud	38.40K Baud	9600 Baud	4800 Baud
0	1	0	4	32.768K Baud	8192 Baud	19.20K Baud	4800 Baud	2400 Baud
0	1	1	8	16.384K Baud	4096 Baud	9600 Baud	2400 Baud	1200 Baud
1	0	0	16	8192 Baud	2048 Baud	4800 Baud	1200 Baud	600 Baud
1	0	1	32	4096 Baud	1024 Baud	2400 Baud	600 Baud	300 Baud
1	1	0	64	2048 Baud	512 Baud	1200 Baud	300 Baud	150 Baud
1	1	1	128	1024 Baud	256 Baud	600 Baud	150 Baud	75 Baud